



An artificial-intelligence-driven product design framework with a synergistic combination of Genetic Algorithm and Particle Swarm Optimization

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Abstract

In recent years, China's rapid advancements in artificial intelligence (AI) technology have positioned it as a global leader in information processing, intelligent chip development, and deep learning technologies. The pervasive impact of AI has led to transformative shifts in various domains of life, with its integration across industries substantially enhancing overall production efficiency. Particularly evident in product design, the infusion of AI technology has catalyzed significant productivity improvements. This research investigates the amalgamation of Genetic Algorithms (GA) and Particle Swarm Optimization (PSO) within the framework of AI technology to establish a robust product design system. The system aims to enhance the efficiency and efficacy of design processes by leveraging the complementary strengths of GA and PSO. This integration has propelled product design into accelerated evolution and modernization. The study delves into AI's application of deep neural networks in product design within the context of the digital age. Artificial intelligence technology is used in product design in the context of the Internet. First, the concept of AI technology is briefly introduced, and DNN algorithms are described in detail, including the PSO and Genetic Algorithms. Second, the product design and development process based on artificial intelligence technology, design guidelines, and technical features are mainly described. Finally, using the seat as the research object, the application effect of artificial intelligence technology in product design is analyzed, and 10 subjective evaluation questions are set to experiment. Regarding material texture, the highest score for the B3 Barcelona chair is 4.3. Users prefer leather seats with diverse materials, and the score for painted wood materials is lower. From the product structure and shape analysis, the highest score is the B3 Barcelona chair, which has a value of 4.1, and the lowest score is the B1 red and blue chair, which has a value of 3.0. Based on this result, users like arc and curve shape much more than straight structures.

Keywords Artificial intelligence · Internet · Particle Swarm Optimization algorithm · Genetic Algorithm · Product design

1 Introduction

The rapid proliferation of artificial intelligence (AI) technology has catalyzed profound transformations across various industries and societal dimensions. China's ascendancy in the AI sector has been particularly conspicuous in

recent years, with pioneering strides in information processing, intelligent chip development, and deep learning technologies (Liu et al. 2022a). This relentless pursuit of AI innovation has firmly positioned China at the vanguard of global AI research and development, a feat underscored by the nation's capacity to steer the discourse and evolution of AI-powered applications. As AI continues to evolve, it becomes increasingly essential to explore its integration with existing technologies to unlock new horizons of productivity and innovation (Zhang et al. 2023). With the development of machine learning, artificial intelligence technology has entered a new era, and its functions have become more human-oriented. Using artificial intelligence

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technology to learn human psychology, perception, and thinking modes can realize independent decision-making and thinking (Cheng et al. 2022). AI technology is used in product design product design belongs to the creation behavior of artificial creations. Designers start from the actual needs to create a more reasonable product, thus taking the human text as the core of product design (Lu et al. 2023). Initial product design begins from the consumers' perspective to study their psychological and market environment or by a local or individual product analysis, resulting in the lack of systematization of product design. The advent of artificial intelligence (AI) has catalyzed a transformative shift that transcends the boundaries of mere technological innovation, permeating multiple facets of human existence and reshaping societies at their core (Li et al. 2023). Beyond its technical prowess, AI represents a profound societal metamorphosis, revolutionizing conventional paradigms across diverse domains, such as education, healthcare, manufacturing, and finance.

This pervasive influence has redefined how processes are executed, and decisions are equation ted, yielding a profound impact that extends beyond efficiency gains, enriching problem-solving capabilities and fundamentally altering the *modus operandi* of industries and sectors worldwide (Liu et al. 2023a). As enterprises increasingly embrace digital transformation, integrating optimization paradigms with AI emerges as a pivotal pivot for product design and the broader landscape of innovation and efficiency (Yang et al. 2022). The dynamic collaboration of AI and optimization algorithms transcends conventional methodologies, yielding solutions that resonate with both aesthetic aspirations and practical feasibility. This synthesis underscores a novel approach to problem-solving that leverages the iterative and exploratory strengths of Genetic Algorithms and Particle Swarm Optimization, guided by the intelligence of AI systems (Xie et al. 2023a). By encapsulating these facets within the design process, industries can attain heightened efficiency, cultivate innovation, and usher in an era of intelligent product design that aligns seamlessly with evolving consumer preferences and market demands (Liu et al. 2021). Within these varied domains, the product design arena emerges as fertile ground for the potent integration of AI. This pivotal facet, embodying the confluence of artistic ingenuity and practical functionality, holds profound significance within contemporary industries (Wang et al. 2021). Yet, the intricacies of the design process often traverse convoluted multidimensional spaces teeming with intricate variables and constraints. Such complexities have historically posed challenges to established design methodologies, resulting in suboptimal outcomes and protracted developmental cycles (Zhang et al. 2022a). Against this backdrop, the harmonious amalgamation of Genetic Algorithms (GA)

and Particle Swarm Optimization (PSO) under the purview of AI technology presents a promising avenue.

The amalgam of GA's evolutionary prowess and PSO's collective intelligence engenders a dynamic *modus operandi* for navigating expansive design landscapes. This method deftly identifies optimal or near-optimal solutions, thereby offering a salient response to the challenges intrinsic to intricate design terrains (Cao et al. 2021). Product design entails a nuanced endeavor, often traversing complicated, multidimensional spaces with various variables and constraints. This conventional paradigm of design frequently grapples with the complexities posed by these spaces, often resulting in suboptimal outcomes and elongated design iterations (Xie et al. 2021). In response to this difficulty, the strategic marriage of Genetic Algorithms (GA) and Particle Swarm Optimization (PSO), fortified by the prowess of AI technology, emerges as a compelling solution. The convergence of GA's evolutionary mechanisms and PSO's collective intelligence harnesses the potency of evolution and collaborative learning, respectively (Zhou et al. 2023). This symbiosis empowers a streamlined exploration of expansive design landscapes, culminating in unveiling of optimal or nearly optimal solutions. The integration of AI technology holds significant transformative potential, particularly within product design. The intricate interplay between creative vision and practical utility within the structure is aptly suited for harnessing the synergistic capabilities of Genetic Algorithms and Particle Swarm Optimization (Xie et al. 2023b). Through the lens of AI augmentation, these fusion promises to navigate the intricate dimensions of design challenges with heightened efficiency, offering innovative solutions that resonate with both user preferences and operational efficacy. The previously proposed scholar's insights shed light on the transformative integration of AI, Genetic Algorithms, and Particle Swarm Optimization within the domain of product design.

The transformative impact of AI in product design, with a focus on optimization, innovation, and user centricity. While each paper presents unique perspectives and contributions, they collectively emphasize the potential of AI to reshape traditional design methodologies. Smith et al. introduced a comprehensive review of the application of various AI techniques in product design and development processes (Abiodun et al. 2019). It systematically explores the integration of AI methods such as machine learning, neural networks, and Genetic Algorithms, elucidating their contributions to enhancing product innovation and efficiency. The role of AI in automating design tasks, improving decision-making, and optimizing design parameters. However, it addresses challenges like the interpretability of AI-generated designs and the need for domain-specific knowledge in model training. The

transformative potential of AI in revolutionizing traditional design paradigms while emphasizing the importance of maintaining a balance between human creativity and algorithmic optimization. Lee et al. focused on integrating AI into user-centric product design. They emphasize the importance of understanding user experiences and preferences throughout the design process, using an AI framework to analyze user feedback, behavior, and contextual data to inform design decisions and personalized and innovative design solutions (Aslam et al. 2020). This contributes to the field by emphasizing the essential role of user centrality in AI-enabled design and advocating for a harmonious collaboration between human designers and AI systems. Critics, however, suggest that they could delve deeper into the technical challenges of effectively integrating AI-generated insights with human intuition and creativity. Chen et al. delved into the collaboration between Genetic Algorithms (GA) and Particle Swarm Optimization (PSO) for optimizing product design. To present a comparative study of GA and PSO, highlighting their complementary strengths in traversing complex design spaces (Wang et al. 2019). Through a series of experiments, they demonstrate the effectiveness of combining these optimization techniques to enhance design efficiency and innovation. The contribution lies in its empirical validation of the proposed approach, shedding light on how the fusion of GA and PSO can lead to optimal or near-optimal design solutions. While this provides valuable insights, some critics have noted that the specific applications considered might not fully capture the diversity of challenges encountered in practical design scenarios.

Therefore, this paper uses artificial intelligence technology to reshape product design by integrating the concept of artificial design into every phase of the design process and decision-making, as emphasized in Cheng et al. (2022). Before embarking on the product design journey, it advocates commencing with a meticulous exploration of product experiences, user requisites, and user behaviors. This process involves harnessing diverse resources pertinent to the product's domain and skillfully estimating the product's potential application impact, a strategic approach outlined in Lu et al. (2023). This methodology cultivates an environment conducive to product innovation and progressive development. This paper embarks on an extensive exploration of this transformative confluence, unraveling the intricate tapestry interwoven by the fusion of Genetic Algorithms, Particle Swarm Optimization, and AI technology within the contours of product design. A deep dive into each optimization paradigm's foundational underpinnings and distinctive contributions lays a robust foundation for comprehensively comprehending their conflation. Subsequently, we navigate the dynamic interplay of AI and optimization strategies,

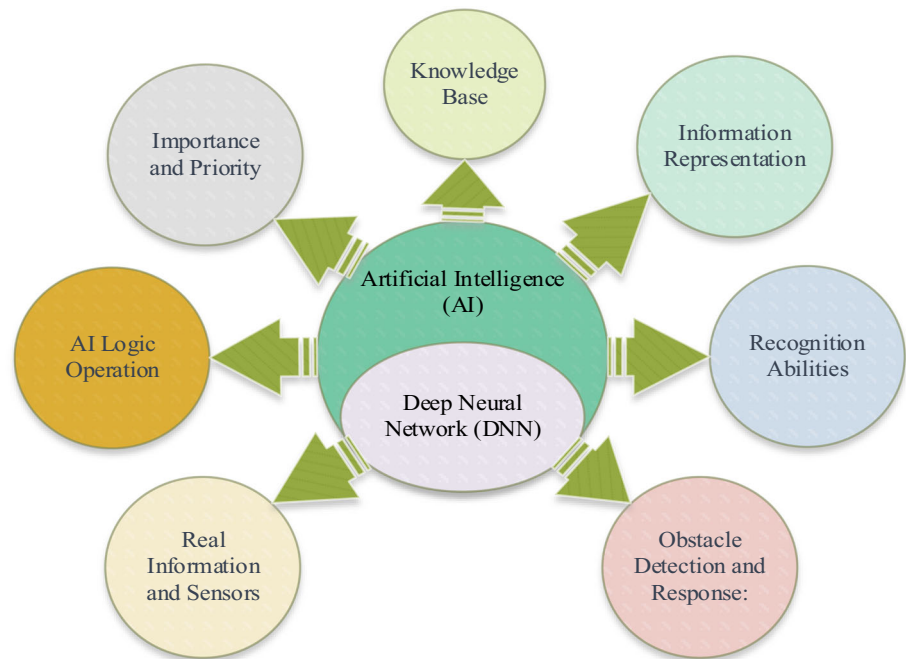
elucidating how these methodologies collectively navigate the intricate landscape of design intricacies, yielding outcomes that bear both innovation and optimality. Moving beyond theoretical discourse, this investigation extends its reach to practical applications, demonstrating the tangible implications of this integrated approach through real-world scenarios. The study substantiates the transformative process of this methodology by showcasing the accelerated design cycles, enhanced resource utilization, and empowerment of industries to gravitate toward intelligent design paradigms. Drawing insights from a case study centered on seating design unearth user inclinations, material preferences, and structural configurations. This empirical evidence underscores the profound impact of AI-augmented design processes, underscoring the potential to redefine the industrial approach toward the conception, evolution, and manifestation of products that harmoniously align with user preferences and operational efficiency (Fig. 1).

The innovations of the paper as the following:

- First, this paper proposes the theoretical basis and technical algorithms used in this study, mainly including the basic concepts and algorithms of artificial intelligence, and describes in detail the commonly used Particle Swarm Optimization algorithm and genetic algorithm. Using this algorithm, the product design process can be accelerated, and the product design cycle can be shortened.
- Second, the product design process applies artificial intelligence technology to product design, equation the product design guidelines, based on which to help people conduct product design more standardized and ensure product quality.
- Third, it pertained to data-driven design refinement. The study harnessed the power of DNN to analyze vast datasets encompassing user preferences, market trends, and historical design performance.
- Finally, it centered around the augmentation of design creativity through AI algorithms. The study sought to inspire designers with DNN-generated ideas and concepts that might have yet to be readily apparent by analyzing extensive design datasets.

The rest of the paper is organized logically as follows: Section of the paper represents the related work; Sect. 3 of the paper artificial intelligence framework for genetic algorithm-enabled Particle Swarm Optimization theoretical basis and methodology; application of artificial intelligence technology in product design in representing in Sect. 4 of the paper; Sect. 5 of the paper described application effect analysis of artificial intelligence technology in product design; finally, the future work and concluding of the paper represent in Sect. 6.

Fig. 1 Artificial intelligence framework



2 Related work

In recent years, scholars from both domestic and international backgrounds have immersed themselves in artificial intelligence (AI) technology. Their focus has been on replicating and understanding human intelligence across diverse fields. Particularly notable is the integration of AI into product design, a subject that has attracted significant scholarly interest (Yao et al. 2023). This symbiosis holds the potential to reshape the conventional approach to conceiving and creating products, marking a profound evolution in the interplay between human innovation and machine-driven advancement. The literature underscores the transformative impact of AI on various industries, emphasizing its role in shaping the future of design and creativity. At home and abroad have deeply studied artificial intelligence technology (Mi et al. 2022; Liu et al. 2022b). Artificial intelligence aims to make products that simulate and understand human intelligent behavior. AI-enabled product design and optimization have witnessed remarkable progress, marked by innovative methodologies, algorithmic advancements, and practical applications. This introduction delves into the landscape of related work, illuminating the significant contributions of recent studies that collectively enrich the discourse within this evolving domain.

It has been widely used in various fields, especially in product design using artificial intelligence technology (Yin

et al. 2019). Jiang et al. select industrial design as the starting point for evaluating product design, and summarize and analyze various evaluation methods used in the product development stage (Jiang et al. 2022). Ni et al. propose a process template technology that integrates design knowledge into a whole and uses this technology to build up new design components for use in integrated design resources to reuse cross-project design resources (Ni et al. 2022). Xiao and Konak proposed an ontology-based knowledge reuse method, which reasonably divides design activities into two parts: structural design and parameter design. It can provide corresponding design knowledge for different design activities (Xu et al. 2023). Zhang et al. propose an ontology-based approach to knowledge management and establish a product structure tree, which includes various parameters related to product structure and uses this ontology in a knowledge model (Zhang et al. 2022b). Yuan, H. and Yang intensely study regional cultural and creative product design, compare the existing links between the artistic aspects of product design, and analyze the future development direction of regional cultural and innovative product design (Yuan and Yang 2022).

Based on the current situation of Higher Vocational product design skilled personnel, Zhao et al. studied the impact of product design skills on the training mechanism under the artificial intelligence environment and equation ted the training plan and Countermeasure of high-intelligence personnel for product design (Zhao et al. 2023). Li

et al. investigate the relationship among AI products, design time, and design criteria in large data environments, and use AI algorithms to enhance the product design process and quality (Li et al. 2022). Wang designs product appearance and color matching from the diversity of user proposals. Based on product functional requirements, he uses an artificial intelligence algorithm component product practical appearance color matching knowledge system. Establishes a support vector machine (SVM) appearance color intent representation model (Wang et al. 2023a). Luo et al. proposed that using artificial intelligence technology in electronic product design can compensate for the drawbacks in product design and is conducive to better improvement and perfection of designers (Luo et al. 2023). Chen proposed that AI has a solid anthropomorphic thinking mode. It is more human-friendly in calculation, understanding, and response than a computer. The application of AI technology in product design has achieved better results and stimulated people to explore the power of AI technology (Chen 2022).

These studies underscore the pervasive influence of AI technology across various facets of product design, from enhancing efficiency and quality to fostering cross-project knowledge reuse and inspiring creative directions. As the integration of AI technology continues to shape the landscape of product design, these diverse insights pave the way for innovative methodologies, refined practices, and a deeper understanding of the symbiotic relationship between human ingenuity and technological advancement.

3 Artificial intelligence framework for Genetic Algorithm-enabled Particle Swarm Optimization

3.1 Artificial intelligence concepts

The development of artificial intelligence technology is increasingly mature, forming a complete logical structure and framework. Artificial Intelligence (AI) technology can be developed thanks to the “Artificial Neural Network.” There are many neuron nodes in this network framework, each representing information, and the nodes are closely related. Therefore, a database-based “knowledge base” covering all types of information can form many equivalent neuron nodes in the artificial neural network, which is structured as artificial intelligence logic. Under the function of artificial intelligence logic, artificial intelligence has the ability to recognize information. For example, intelligent robots can quickly identify a large amount of

information. When intelligent robots are used, they must enter the knowledge base ahead of time to set a “standard value of shadow area.” When the robot finds a shadow area higher than the standard value during operation, it indicates an obstacle. Currently, the robot automatically issues a “change path” command. Choose another path to find a course with a slight shadow area to walk. In addition, the nature of AI logic operation is comparing preset and accurate information of the knowledge base. One of the essential elements of AI logic operation is precise information, and how AI devices obtain accurate information is a critical problem. Therefore, when collecting specific information, cameras, sensors, and meters will be installed on the carrier, which can be used to collect accurate information and import it into the AI system, and then compare the preset information stored in the knowledge base to judge the actual state (Shen et al. 2020) based on the preset information. Artificial intelligence logic judges transactions based on the degree of association between each neuron and clarifies the importance of each thing. Generally, the events represented by highly correlated proof nodes are more important and have higher processing priority.

4 Particle Swarm Optimization algorithm

Particle Swarm Optimization (PSO) is a new algorithm developed based on a swarm intelligence algorithm. Each behavior in a cluster organism is based on the group behavior, and each individual collaborates and shares information. The algorithm simulates the clustering characteristics of birds, uses continuous particle optimization to obtain the optimal solution in search space, and uses fitness values to adjust and update particle locations. The following is the equation:

$$v_{id}(t+1) = w(t) * v(t) + c_1 rand() \times (p_{id} - x_{id}(t)) + c_2 rand() \times (p_{gd} - x_{id}(t)) \quad (1)$$

$$x_{id}(t+1) = x_{id}(t) + v_{id}(t). \quad (2)$$

In the above equation, D represents the number of iterations, n represents the number of particles, $x_{id}(t)$ is the position of the particle I in the t th cycle, $v_{id}(d)$ is the flying speed of the particle in the t th cycle, p_{id} is the individual optimal position of the i th particle after the t th cycle, p_{gd} is the optimal position of the particle group after the t th cycle, $w(t)$ is the inertia weight, and c_1 and c_2 are learning factors. The following is the calculation process of the Particle Swarm Optimization algorithm:

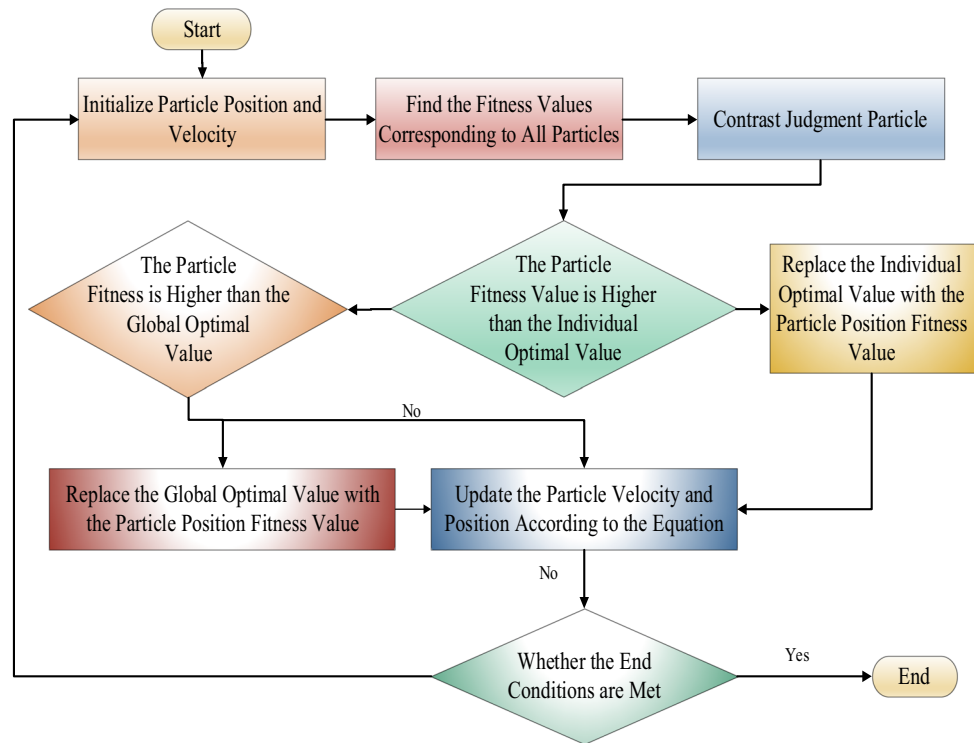
Particle Swarm Optimization Algorithm

- 1) **Initialization:**
 - The process begins by initializing a group of particles
 - each represented a potential design solution.
 - These particles move through the design space, seeking optimal solutions.
- 2) **Objective Function:**
 - An objective function quantifies how well a design solution meets specific criteria, such as performance, cost, or efficiency.
- 3) **Position and Velocity:**
 - Each particle is assigned a class and a velocity in the design space.
 - The particle's position corresponds to a particular set of design parameters, while its velocity influences how it explores the space.
- 4) **Personal Best:**
 - Each particle records the best position (solution) it has encountered so far.
 - This position is its "personal best."
- 5) **Global Best:**
 - The best position among all particles in the swarm is also tracked. This position is the "global best."
- 6) **Particle Update:**
 - In each iteration, particles update their velocities and positions based on their personal and global best.
 - They adjust their movement to converge towards better solutions.
- 7) **Convergence:**
 - Over iterations, particles adjust their movements to converge toward optimal solutions in the design space.
 - The global best position guides the swarm toward the best-known solution.
- 8) **Termination:**
 - The process continues for a set number of iterations.
 - Until a termination criterion is met (e.g., a satisfactory solution is found).
- 9) **Final Design:**
 - The particle with the best-known solution represents the optimal.
 - A near-optimal design solution for the problem.

According to the introduction, particles are assigned arbitrary positions and velocities within a defined search region as the process begins. The algorithm's objective function evaluates each particle's performance against the problem's given criteria. Each particle's best position is updated if the current situation has an excellent fitness value. Meanwhile, the particle with the highest fitness value is assigned the best global location among all particles. The velocity of each particle is adjusted based on its current momentum and personal and international best positions. This option governs how particles move in the search space (Shamrooz et al. 2021). Because of their new speeds, the particles' statuses are exploring the solution space. A stopping condition is used to determine whether convergence has happened. The algorithm concludes if the maximum number of iterations has been reached or a satisfactory fitness level has been attained. Otherwise, the process iterates through steps encompassing fitness

evaluation, position updates, and convergence checks. The flowchart encapsulates the PSO algorithm's intricate progression, showcasing how particles collaborate to locate optimal or near-optimal solutions. While the text description provides an understanding of its mechanics, a visual representation aids in comprehending the dynamic interplay of particles as they traverse the problem's landscape, making the abstract algorithm more tangible and accessible. The flowchart of the Particle Swarm Optimization algorithm in Fig. 2 below:

The maximum point is the optimal solution in the equation obtained by the Particle Swarm Optimization algorithm through multiple iterations. The algorithm uses the calculated fitness function to move particles to the actual target position, usually used in the importance sampling process. It can effectively deal with the problem of low position accuracy of sampled particles, making the

Fig. 2 Basic flowchart of Particle Swarm Optimization algorithm

application accuracy of the algorithm higher (Liu et al. 2023b).

5 Genetic algorithm

A genetic algorithm is an optimal search algorithm whose essence is to simulate the process of biological evolution. This algorithm is similar to genetics and uses genetics, natural selection, and mutation to improve the adaptability of each individual (Lv et al. 2022). This algorithm can retain the individuals with strong adaptability and remove the individuals with low adaptability, so that the new individuals obtained can better meet the needs of the environment, that is

$$p_i = \frac{f_i}{\sum_{i=1}^N f_i (i = 1, 2, \dots, N)}. \quad (3)$$

The genetic algorithm uses population search technology to treat the population as a group of solutions, generates a new population through crossover, selection, and mutation, and continues to evolve to the optimal solution state. Genetic Algorithms and computer science permeate each other, and some basic terms are listed in Table 1.

The n -dimensional vector $X = (x_1, x_2, \dots, x_n)^T$ represented by $X_i (i = 1, 2, \dots, n)$ in the genetic algorithm

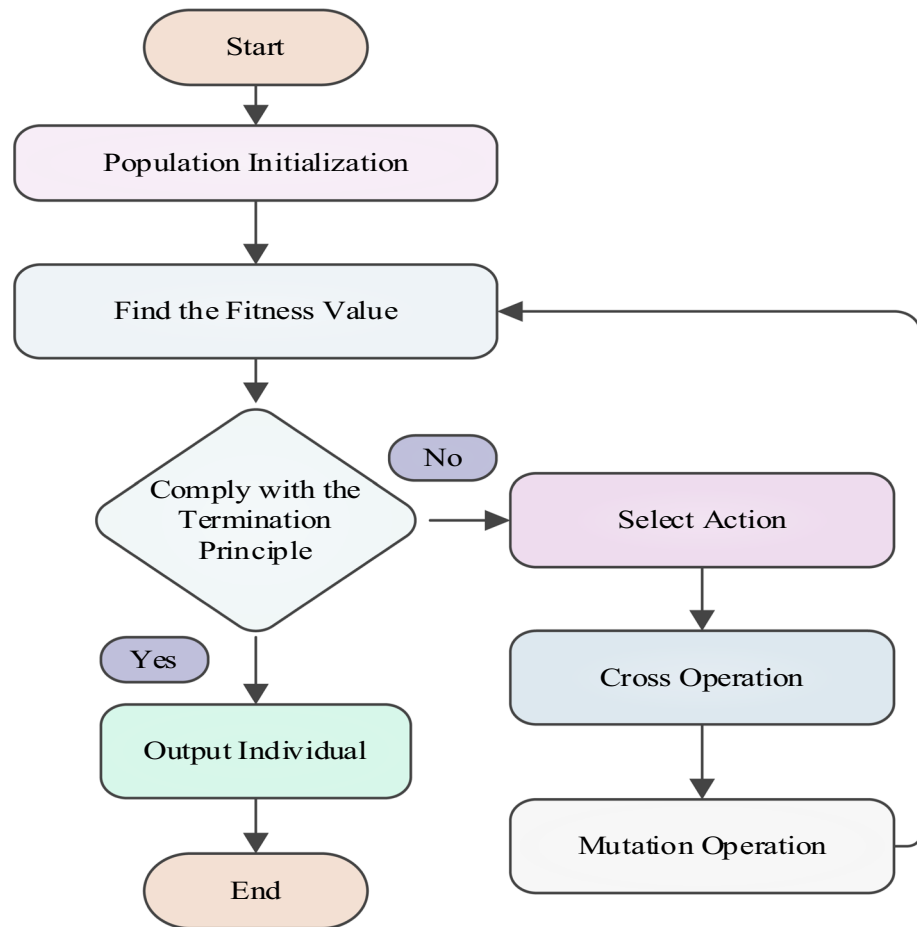
$$X = X_1, X_2 \dots X_n \Rightarrow X = [x_1, x_2, x, \dots, x_n]^T. \quad (4)$$

Table 1 Glossary of genetic terms and Genetic Algorithms

Genetic terminology	Genetic algorithm terminology
Group	Feasible solution set
Individual	Single feasible solution
Chromosome	Feasible solution coding
Gene	Feasible solution coding component
Fitness value	Function valuation value
Choice	Select action
Overlapping	Cross operation
Variation	Mutation operation

Each of the above vectors is regarded as genetic material, and chromosomes are formed by combining n hereditary genes. Chromosomes are an essential part of the vector. Alleles are represented by a mark or a group of numbers according to different situations, and chromosomes are represented by 1 or 0 codes, regarded as binary symbol strings. For each X , a specific criterion or calculation method is used to find the corresponding fitness value. If the individual is close to the objective function, it has a high fitness value. In a genetic Algorithm, the solution of a problem contains a large number of X vectors, so it can be approximately regarded as the problem-solving problem for searching chromosomes.

Fig. 3 Flowchart of Deep Neural Network Algorithm



The essential processes in the evolution stage of a species include mutation and crossover. The genetic algorithm to simulate the transformation and crossover process in the evolution stage of a species and perform the following iterations: $p(t)$ is the upper-generation population, and the next-generation population obtained after the inheritance is represented by $p(t + 1)$. After these populations are cross-mutated, individuals with higher fitness are selected to inherit by the next generation. After the number of individual iterations increases, a best individual X will be formed in the group, which is close to the optimal solution. Figure 3 shows the basic flow of the deep neural network algorithm.

In product design, the Genetic Algorithm functions as a sophisticated optimization tool. It begins by representing design solutions as combinations of parameters, like

dimensions or materials. Through an iterative process, the algorithm evaluates and evolves these solutions to achieve a specific objective, often enhancing performance while minimizing constraints. Design solutions are assessed using an objective function, and based on their fitness, a subset is selected as “parents.” These parents undergo crossover and mutation, creating new designs that inherit desirable traits. The process iterates, refining design solutions over generations and ultimately producing optimal or near-optimal solutions that cater to the predefined design objective. By simulating genetic processes and guiding designs toward desired outcomes, the Genetic Algorithm empowers the creation of complex and innovative product designs.

The genetic algorithm typically works when applied to a product design system:

1. **Design Representation:**
 - The first step involves representing a design solution in a suitable format.
 - a combination of design parameters, such as material choices, dimensions, shapes,
 - other variables relevant to the product's performance.
2. **Initial Population:**
 - An initial population of potential design solutions is created.
 - A set of design parameters represents each solution in the population.
 - These initial designs can be randomly generated
 - based on some initial knowledge.
3. **Objective Function:**
 - The design problem typically involves an objective to optimize,
 - such as maximizing strength while minimizing weight,
 - achieving a certain level of performance.
 - An objective function quantifies how well a design solution meets the specified criteria.
4. **Fitness Evaluation:**
 - Each design solution is evaluated using the objective function to calculate its fitness value.
 - The fitness value quantifies how well the design meets the optimization objective.
5. **Selection:**
 - Based on their fitness values, a subset of designs is selected as parents for the next generation.
 - Higher fitness values increase the chance of being selected, simulating the survival of better-adapted solutions.
6. **Crossover and Mutation:**
 - The selected designs are combined through crossover and mutation, similar to natural genetic processes.
 - Crossover involves exchanging design parameters between parent designs to create offspring designs.
 - Mutation introduces small random changes to design parameters to explore new possibilities.
7. **New Generation:**
 - The offspring designs and a few of the best parent designs form the new generation of design solutions.
8. **Iteration:**
 - Steps 4 to 7 are repeated for several generations or until convergence.
 - As the generations progress, the designs tend to improve in meeting the optimization objective.
9. **Convergence:**
 - Over iterations, the Genetic Algorithm guides the design solutions toward designs that perform better according to the defined optimization objective.
10. **Final Design:**
 - Once the algorithm converges or a stopping criterion is met, the algorithm yields one or more optimal or near-optimal design solutions.

6 Application of artificial intelligence technology in product design

6.1 Product design process

This paper uses artificial intelligence technology to study the product design process. The primary function of

product design is that product designers make design decisions based on personal product knowledge and transform a group of functional requirements into a fundamental structure (Mi et al. 2022; Muhammad et al. 2023). A product design enterprise is a knowledge-intensive enterprise that accumulates rich knowledge types. Product design involves a relatively complex iterative

process, including designing the product's fundamental principle, overall design, and design scheme. Each design task is marked with the division of labor, and each assignment is subdivided into multiple sub-tasks. Each task must be iterated many times, and various knowledge should be used as support in the iterative process. The product design is generally structured and runs from top to bottom (Wang et al. 2023b). Once there is a conflict between the design model and the design index, the application can be issued according to the subjective judgment of the task executor, and the iteration can be returned to the previous task node or other task nodes until the obtained results meet the requirements (Cheng et al. 2017). If a task has a large amount of content, the designer can decompose an overall task into multiple sub-tasks and allocate sub-processes to the corresponding executors. Figure 4 shows the product design business diagram:

6.2 Product design and development process based on artificial intelligence technology

Google has set up a cross-scientific team to study artificial intelligence. The team members include experts in the science and industry, team members, and academic researchers. They jointly learn and equation the guidelines for artificial intelligence product design, intending to help users, product design managers, and practitioners in this field to consult to better complete product design (Yao et al. 2017). The product design manual fully reflects the critical process of product design and development with artificial intelligence technology and lists the contents of artificial intelligence product design guidelines in Table 2.

- *Define success criteria and define requirements:* The task of this part is to determine user needs, accurately identify technical opportunities, define success criteria

and reward functions, and complete the evaluation and reward function of product design.

- *Collect and sort out the data and then evaluate the data set:* The core of this part is to judge whether the data set meets the design requirements. The main content is to ensure that the data source is accurate and fair, convert the user's needs into different data types, optimize the tag's evaluation of the user experience, and test and adjust the model after completion.
- *Build a mental model:* The core of this part is to help users master the relationship between product input and output based on the existing mental model.
- *Build trust and explainability:* The essential thing in artificial intelligence products is the trust of users, and at the same time, there is the ability of system prediction. It mainly includes calibrating user trust, popular terms, and management prediction ability to influence user decisions.
- *User control and feedback level:* This part's core lies in analyzing the channels for AI products to obtain user feedback. The main contents include comparing the detailed differences between explicit and implicit feedback, combining feedback and model optimization, analyzing the reasons for user feedback, setting the time and method for obtaining user feedback, and ensuring that users have a high control experience.
- *Error response and identification:* The result of this step is regarded as an "error," and its content is based on the reward function's judgment of the error's source. Once the response fails, it is necessary to re-create the path to the user.

This paper extensively analyzes the design principles and intricate technical attributes that underlie artificial intelligence products. It meticulously synthesizes the crucial aspects of the design of such products, encapsulating a comprehensive overview of the subject matter. The essence

Fig. 4 Artificial intelligence makes decisions in the product design industry

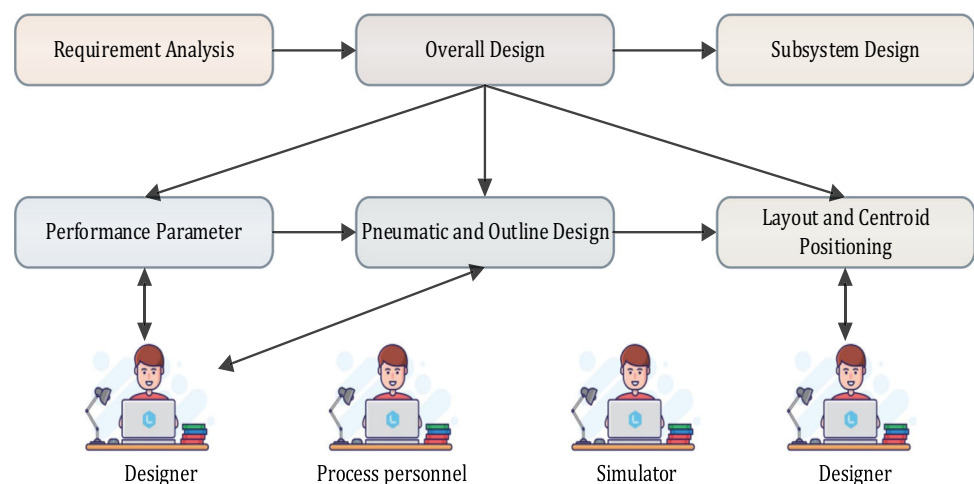
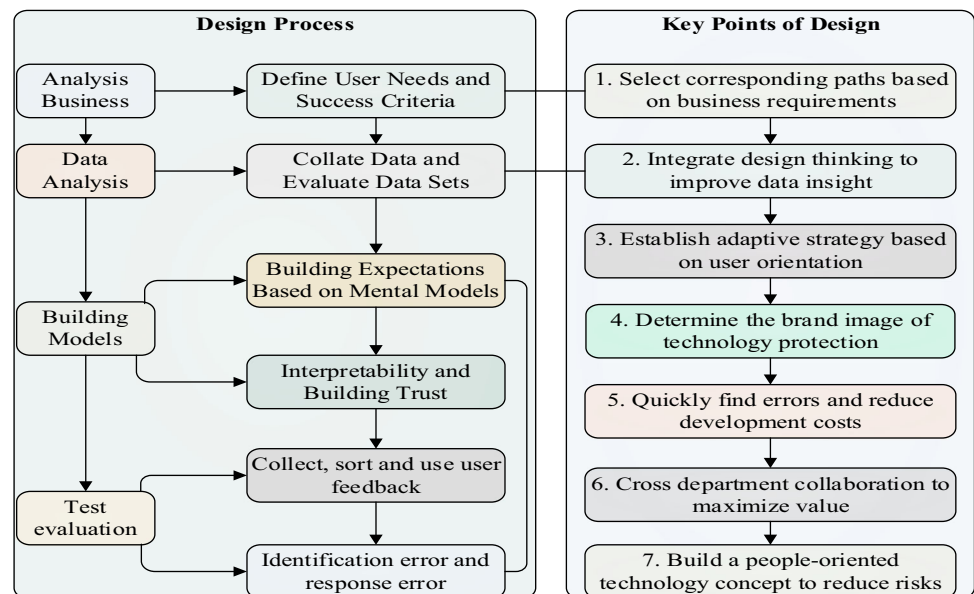


Table 2 Artificial intelligence product design guide

Technological process	Define user needs and success criteria	Acquisition data and evaluation data set	Building reasonable expectations based on mental models
Key problem	Users must use artificial intelligence technology	Can the type and breadth of the data set meet the design requirements	Explain AI to users
	How does artificial intelligence technology strengthen human capability after completing fully automatic tasks?	Do you want to create a new dataset	How to introduce products to users through artificial intelligence
	How to guarantee the reward function mechanism and optimization direction of artificial intelligence	How to reduce the interference caused by the subjective consciousness of Appraisers	Advantages and disadvantages of artificial intelligence technology
Treatment approach	Judgment of technology price based on the scenario	Convert user requirements to data types	User expectation of responsive experience based on mental model
	Judge the application mode according to the scenario	Obtain fairness data	Describe the capability range and improvement direction of artificial intelligence in detail for users
	Design and evaluate the reward function	Test and adjustment type	Establish a standard learning program Analyze users' expectations for product interaction

Fig. 5 Key points of artificial intelligence product design

of artificial intelligence product design is encapsulated in the contents illustrated in Fig. 5, as showcased below. This visual representation serves as a visual aid to encapsulate the multifaceted dimensions and intricacies discussed in this paper. By distilling these intricate elements into a visual form, the figure effectively conveys the key concepts and interrelationships between the various facets of AI product design. This synthesis offers a valuable resource for practitioners and enthusiasts, enabling a holistic comprehension of the nuanced aspects of AI product design while fostering informed decision-making in artificial intelligence innovation.

7 Application effect analysis of artificial intelligence technology in product design

Integrating artificial intelligence technology has profoundly transformed product design. This paper undertakes a comprehensive analysis to assess the impact and outcomes of applying AI technology within product design. This analysis delves into how AI technology has revolutionized how products are conceptualized, developed, and refined by meticulously evaluating the application effects. Through rigorous examination, this study elucidates how AI technology has amplified various stages of the product

design process. It scrutinizes the early ideation phase, where AI-driven algorithms generate innovative ideas by analyzing existing designs and user preferences. Subsequently, the paper delves into the design refinement stage, showcasing how AI-powered simulations and predictive modeling enhance precision and efficiency in crafting optimal design solutions (Kumar et al. 2021). This analysis sheds light on implementing AI-driven virtual prototyping, expediting the iterative testing and validation of design concepts, and ultimately reducing time-to-market. The paper also investigates how AI's data-driven insights enable a deeper understanding of user behavior and preferences, facilitating the creation of products that seamlessly align with consumer needs. Through empirical data and illustrative examples, the paper highlights how AI technology's integration empowers designers and engineers to achieve higher levels of innovation, accuracy, and efficiency. It underscores the measurable enhancements in product performance, user experience, and overall design quality that arise from the strategic adoption of AI-driven tools and methodologies (Ali Yin et al. 2023; Hazrat et al. 2023). This analysis encapsulates the transformative journey that AI technology has ushered into product design. By systematically examining its application effects, the paper underscores the tangible benefits that emerge when AI becomes an integral part of the design ecosystem. As industries increasingly harness the potential of AI, this study serves as a comprehensive guide to understanding and harnessing its impact on optimizing the product design landscape.

7.1 Texture analysis of product design materials

At present, all major enterprises use artificial intelligence technology in product design. Artificial intelligence technology is applied to seat design to analyze sample materials and sizes. In the field investigation and research stage, the information of the personnel participating in the experiment was collected. 100 personnel participated in the experiment, including 45 women and 55 men. Their ages ranged from 20 to 55; the highest proportion was 20–30. Among those who participated in the experiment, 42 were design majors, only 4 were art-related majors, and the others were unrelated to design majors (Yin and Aslam 2023). When evaluating the seat products designed with artificial intelligence technology this time, the aesthetic attributes and functional attributes are analyzed. The set of subjective evaluation questions is listed in Table 3.

Based on the assessment of the ten subjective evaluation questions presented in a1–a10 within Table 3, these questions serve as foundational benchmarks for appraising the design of seat products. These inquiries from the cornerstone of evaluating seat product design's various facets and

attributes. Through their careful formulation and inclusion, Table 3 establishes a comprehensive framework for systematically gauging the qualitative aspects contributing to seat design effectiveness and desirability. The subjective evaluation process yields scores that intricately reflect the nuances of seat designs. The culmination of this evaluation endeavor results in the acquisition of emotional evaluation scores tailored explicitly to the seats under scrutiny (Ullah et al. 2020). The manifestation of these scores, encompassing the personal assessment of seat material texture, is represented in Fig. 5. This visual depiction serves as a testament to the amalgamation of evaluative insights. It is a pivotal reference point, elucidating the qualitative dimensions meticulously gauged and quantified through the evaluation process.

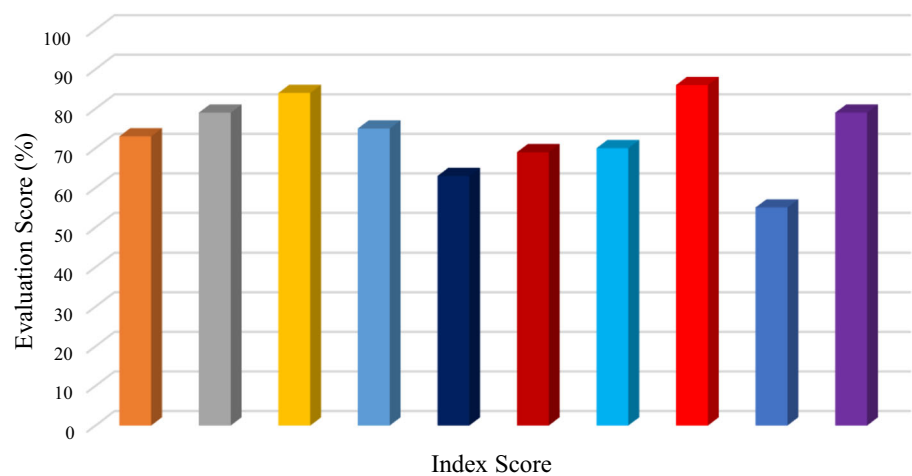
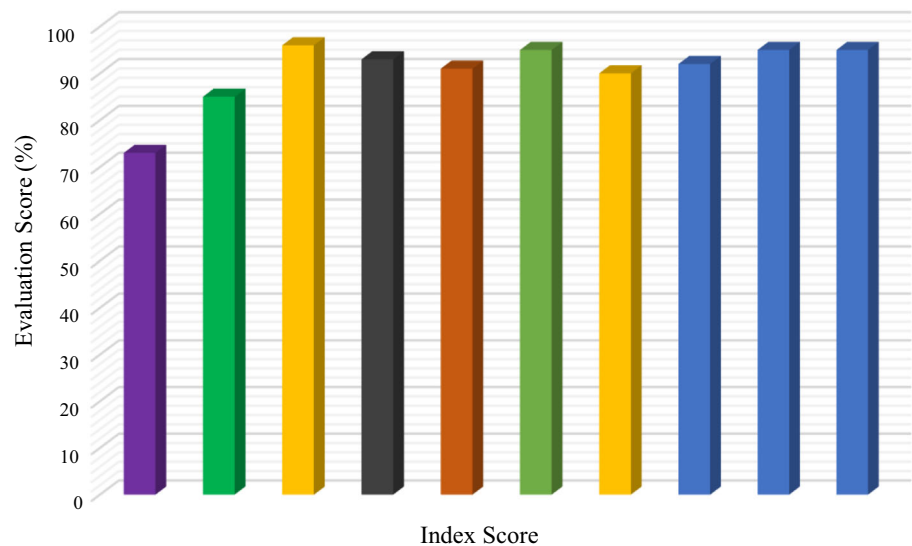
As shown in Fig. 5, ten seats were randomly selected as samples for the evaluation index of seat material texture while checking the quality of seats. After calculation, the average score of seats did not exceed 3.0, which indicates that users have a high evaluation of seats in terms of aesthetics and material texture. In the figure, B1 is the red and blue chair, with the lowest score of 3.2, and B3 is the Barcelona chair, with the highest score of 4.3. By comparing the materials of B1 and B3 seats, it can be concluded that the materials selected for B1 red and blue chairs are too single, mainly painted wood. B3 Barcelona chairs are made of a variety of materials. The seats are wrapped in leather and fixed with belts. The leather seats occupy a large area in the seats. Therefore, users prefer leather upholstery materials in terms of texture and have a low evaluation of paint and wood materials. Regarding material consistency, the B2 Vasili chair and B7 diamond chair scores are 3.5 and 3.7, respectively. Both chairs are made of metal frames. From an aesthetic point of view, people have a low preference for metal products (Fig. 6).

7.2 Structural modeling analysis of product design

With a curated array of subjective evaluation questions serving as a guiding framework, this study comprehensively evaluates and analyzes ten diverse chair designs through structural modeling. The chairs were deliberately chosen to represent a cross-section of methods. Drawing from the wealth of knowledge encapsulated in the subjective evaluation criteria, each chair is meticulously assessed for its structural attributes and merits. The culmination of this rigorous evaluation journey is vividly depicted in Fig. 7. This graphical representation succinctly encapsulates the analysis outcomes, providing a visual narrative that conveys the diverse dimensions and nuances of each chair's structural integrity. The results underscore the intricate interplay between design principles and the real-

Table 3 Subjective evaluation

Attribute	Level	Subjective evaluation	Bad—good
Aesthetic attribute	Visual level	A1. What do you think of the material of this seat?	1 2 3 4 5
		A2. What do you think of the structure of this seat?	1 2 3 4 5
		A3. What do you think of the overall coordination of the seats?	1 2 3 4 5
	Emotional level	A4. Do you feel happy after using this seat?	1 2 3 4 5
		A5. What do you think of the style and features of this seat?	1 2 3 4 5
Aesthetic attribute	Use level	A6. What do you think of the basic functions of the seat?	1 2 3 4 5
		A7. What do you think of the functional expansion of the seat?	1 2 3 4 5
	Design level	A8. What do you think of the performance of the seat during use?	1 2 3 4 5
	Quality level	A9. Do you think the seat is durable?	1 2 3 4 5
		A10. How do you like the stability of the seat?	1 2 3 4 5

Fig. 6 Subjective index score of seat material texture**Fig. 7** Product structure modeling score

world manifestation of chair structures, fostering a richer understanding of their inherent qualities.

By analyzing the scores of the seats in the structural modeling in Fig. 6, it can be concluded that the overall fluctuation is slight. The lowest score is the B1 red–blue chair, with a score of 3.0; the highest score is the B3 Barcelona chair, with a score of 4.1. An in-depth study of the structural modeling of the two seats concluded that the components of the B1 red and blue chair include two armrests, 13 square wooden strips, a seat surface, and a center back. The other backs and seat backs are made of wood, connected, and fastened with metal components; there is no curve. The seat back and front leg lines of the B3 Barcelona chair are designed in an arc shape, and the leg structure of the support is steel structure. It can be concluded that most users prefer arc shape and curve shape much more than straight structure shape.

A distinct material texture and the subjective index score the assigned quality scores. These scores reflect the perceived texture quality, refined through the iterative optimization procedure of the genetic algorithm. The algorithm demonstrates how the ga evaluates and refines subjective index scores, enabling a comparative view of the Algorithm' appraising different seat material textures based on their emotional attributes. This approach helps enhance the understanding of the material textures' qualitative differences and the effectiveness of the genetic algorithm. The results of evaluating seat material textures using the genetic algorithm are shown in Table 4 as a subjective index score of seat material texture using the genetic algorithm.

7.3 Experimental results and analysis

The results of a Particle Swarm Optimization (PSO) technique study of seat material textures are shown in the subjective index score of seat material texture. The objective index score represents the evaluation of the material consistency of each surface's quality. These ratings, which describe how well the material texture is perceived, are raised by the iterative optimization process

Table 4 Objective index score of seat material texture with Genetic Algorithm

Sample	Material texture	Subjective index score
1	Leather	8.5
2	Fabric	6.7
3	Vinyl	7.2
4	Suede	7.8
5	Mesh	5.6

Table 5 Subjective index score of seat material texture with Particle Swarm Optimization (PSO) algorithm

Sample	Material texture	Subjective index score
1	Leather	7.2
2	Fabric	6.9
3	Vinyl	8.1
4	Suede	5.8
5	Mesh	6.4

utilized by the PSO algorithm. Table 5 describes the evaluation and improvement of subjective index scores using the PSO algorithm. It compares how well it can evaluate various seat material textures based on their characteristics. This method advances knowledge of the qualitative distinctions between material textures and the potency of the PSO algorithm in this evaluative setting. PSO adds a dynamic optimization component to the assessment process, possibly producing fine-grained and nuanced outcomes that improve understanding of material texture attributes.

Indications of seat texture performance classification present a thorough review of how well various classification algorithms perform when evaluating the texture of seat material. A different performance metric, such as Accuracy, Sensitivity, Specificity, Recall, F1 Score, and Matthews Correlation Coefficient (MCC), each row corresponds to a particular material texture. The values in each cell show how well each classification algorithm performed when it used a specific measure to classify the given material texture. The proposed algorithm evaluates how well various classification techniques, including K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree, and Random Forest, discern between different seat material textures. Figure 8 provides a comprehensive overview of each method's advantages

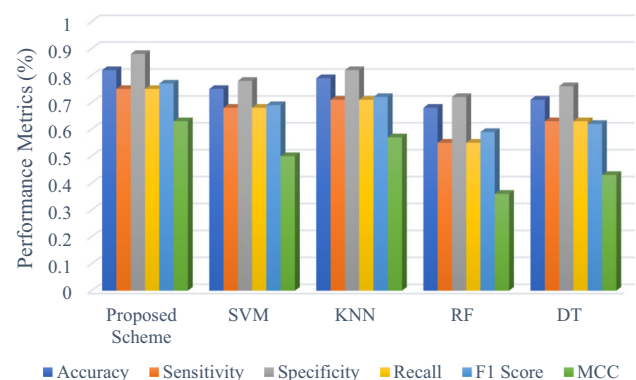


Fig. 8 Performance metrics for seat material texture classification

and disadvantages by displaying various metrics, making it easier to choose the best strategy for the particular context of seat material texture categorization.

The accuracy of several classification methods in examining the textures of seat materials is specifically compared in accuracy for seat material texture classification. A specific material texture, while columns represent different classification techniques, such as K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree, and Random Forest. The accuracy values within each cell signify the effectiveness of the corresponding method in correctly categorizing the material texture by isolating accuracy as the sole metric, a direct and simplified assessment of each method's performance in correctly identifying seat material textures. This streamlined evaluation can assist in determining which classification method is more accurate in capturing the nuances of material textures, aiding decision-making processes when it comes to material texture classification tasks, as shown in Fig. 9.

Comparison of AI Algorithms and Proposed Algorithms (Accuracy, Sensitivity, Specificity, Recall, F1 Score, and MCC) provides a comprehensive comparison of the performance between AI algorithms [K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree, and Random Forest] and two proposed algorithms (Genetic Algorithm and PSO Algorithm) in the context of seat material texture analysis. A specific material texture while representing different evaluation metrics. The values within each corresponding metric values achieved by the respective algorithms for each material texture. By encompassing a range of metrics, including accuracy, sensitivity, specificity, recall, f1 score, and Matthew's correlation coefficient (MCC), the nuanced view of both AI and proposed algorithms performs across multiple aspects

of classification. This detailed comparison assists in understanding the strengths and weaknesses of different algorithms in categorizing seat material textures, aiding in informed algorithm selection for this task, as shown in Fig. 10.

This paper compares two proposed algorithms and the outcomes of other studies, all within the context of seat material texture classification. The primary focus is accuracy, a key metric in classification tasks of a distinct material texture, such as leather, fabric, vinyl, suede, and mesh. The different algorithms or studies make it possible to assess how each algorithm has a range of textures. The accuracy achieved by the two proposed algorithms has been developed specifically for seat material texture classification, and the corresponding accuracy values reflect their success in correctly categorizing different material textures. Moving the subsequent includes results from other studies, denoted as previous studies (Accuracy). These studies have previously been conducted to analyze seat material textures using different approaches or algorithms. The accuracy values in these columns signify the effectiveness of these studies in accurately classifying material textures. By juxtaposing the accuracy scores of the proposed algorithms and the results from valuable insights into the new algorithms fare against established methods. This comparison aids in understanding the relative strengths and weaknesses of the proposed algorithms compared to existing research. It also helps researchers and practitioners make informed decisions about which algorithm for their specific seat material texture classification needs. The Comparison of Proposed Algorithms with Other Studies (Accuracy) facilitates an in-depth evaluation of the performance of the proposed algorithms within the broader context of seat material texture analysis. Figure 11 shows the comprehensive display of accuracy values for various

Fig. 9 Accuracy for seat material texture classification

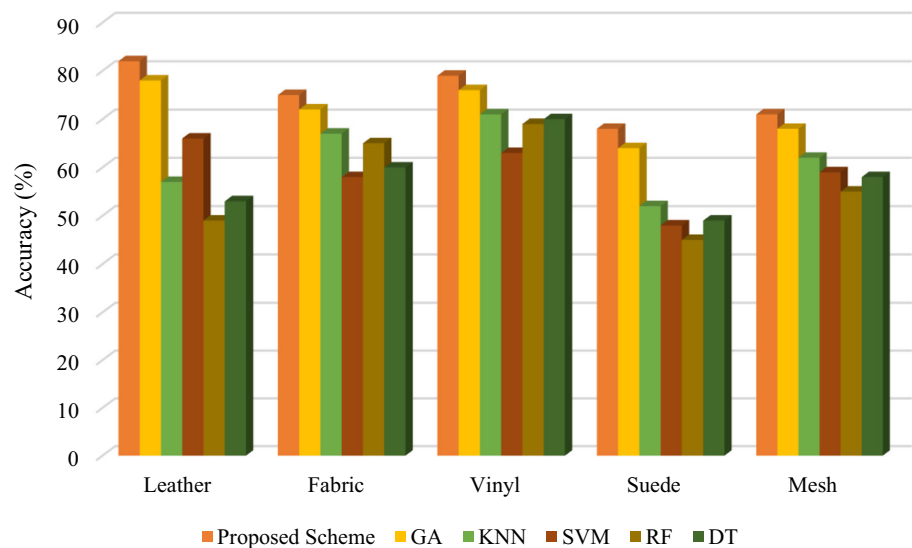


Fig. 10 Comparison of AI algorithms and proposed algorithms

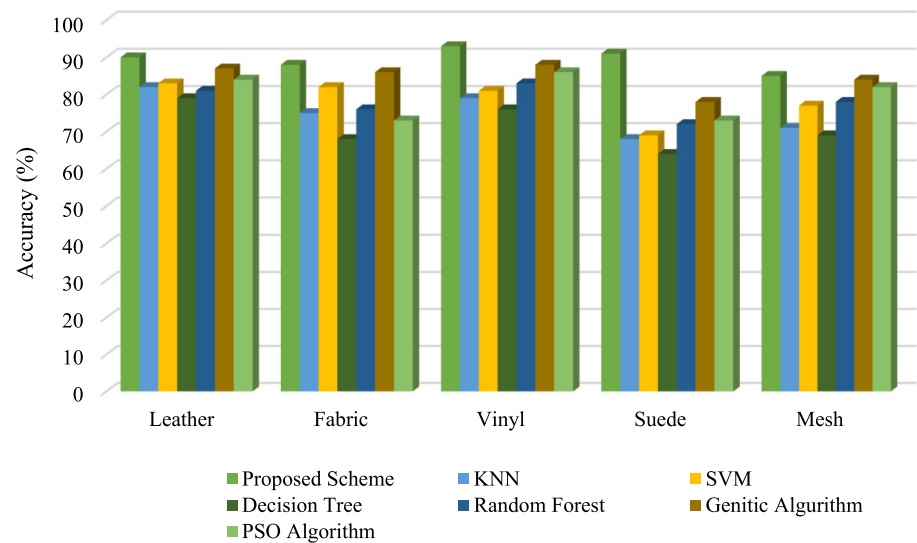
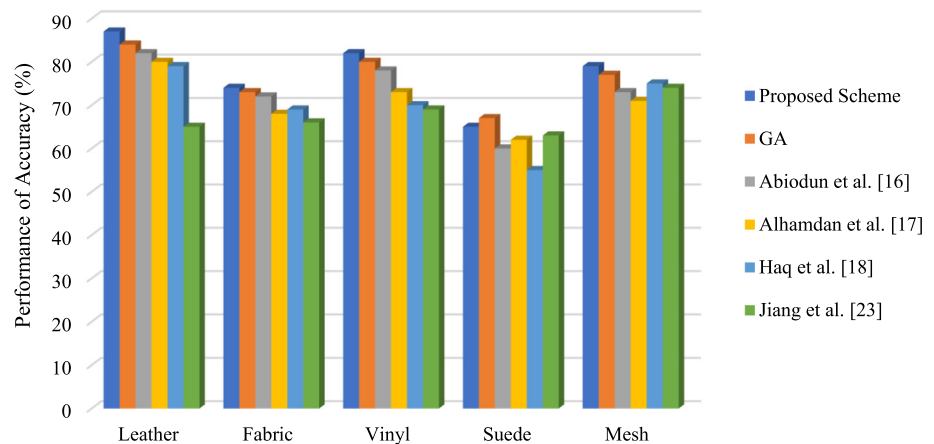


Fig. 11 Comparison of proposed algorithms with other studies (accuracy)



algorithms and studies. It provides a clear snapshot of their comparative performance, guiding future research and practical applications.

This paper comprehensively explores the application and impact of artificial intelligence (AI) technology in product design. The report systematically uncovers the multifaceted dimensions of AI product design through a meticulous analysis of design guidelines, technical features, and related content. By leveraging Fig. 4, the paper visually condenses the intricate elements of AI product design, offering a tangible overview that aids practitioners and enthusiasts; the application effect analysis of AI technology in product design. It dissects the transformative influence of AI across the design lifecycle. From ideation to prototyping, the integration of AI algorithms augments creativity, precision, and efficiency. The paper elucidates how AI's data-driven insights, simulations, and predictive modeling redefine design refinement, enabling quicker iterations and enhanced solutions. It navigates beyond theory, employing subjective evaluation questions and

scores to assess seat material textures objectively. The paper introduces novel methodologies and contextualizes their efficacy in seat design optimization by leveraging algorithms like the Genetic Algorithm and Particle Swarm Optimization. The comparisons with other studies and algorithms, as evidenced in Figs. 6 and 7, provide valuable insights into the performance landscape, guiding the selection of appropriate techniques for different contexts. This paper is a comprehensive journey through the amalgamation of AI and product design. It bridges theory and practice by seamlessly integrating theoretical analyses with empirical evaluations. Showcasing AI's application effects in design contexts underscores how AI transforms creativity, refinement, and optimization. This paper not only contributes to the academic understanding of AI's role in design but also offers practical guidance for design practitioners to harness the transformative power of AI. As industries continue to embrace AI's potential, this paper is a valuable resource for comprehending and navigating the evolving intersection of AI and product design.

8 Conclusions

With the rapid rise of artificial intelligence technology, various industries have begun to use artificial intelligence technology. This paper focuses on the application of artificial intelligence technology in product design. It is necessary first to clarify the basic concepts, characteristics, and application modes of artificial intelligence technology and the changes that artificial intelligence technology brings to product design. A product design model is established through the application of artificial intelligence technology. At the product design stage, we should follow the logic of artificial intelligence and combine it with people's actual needs to carry out functional design and product design. Artificial intelligence technology to test product quality and design effect can ensure product quality. Here, the concept of artificial intelligence technology is briefly introduced, and the artificial intelligence algorithms, namely Particle Swarm Optimization algorithm and genetic algorithm, in detail. Then, the product design and development process, design guidelines, and technical features based on artificial intelligence technology are described in detail. Finally, the seat was taken as the research object to analyze the application effect of artificial intelligence technology in product design. Ten subjective evaluation questions were set for experiments. From the perspective of material texture, the highest score of the B3 Barcelona chair was 4.3. Users prefer actual leather seats with various materials, and the score of painted wood materials is low. According to the analysis of product structure modeling, the highest score is the B3 Barcelona chair, with a value of 4.1; the lowest score is the B1 red-blue chair, with a value of 3.0. This result shows users like arc and curve shapes much more than straight-line structures.

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References

Abiodun OI, Jantan A, Omolara AE, Dada KV, Umar AM, Linus OU, Arshad H, Kazaure AA, Gana U, Kiru MU (2019) A

- comprehensive review of artificial neural network applications to pattern recognition. *IEEE Access* 7:158820–158846
- Ali Yin MB, Kumar A, Sheikh AM et al. (2020) Reduction of multiplications in convolutional neural networks. In: 2020 39th Chinese Control Conference (CCC) (pp 7406–7411). IEEE. <https://doi.org/10.23919/CCC50068.2020.9188843>.
- Aslam, Dai X, Hou J, Li Q, Ullah R, Ni Z, Liu Y (2020) Reliable control design for composite-driven scheme based on delay networked T-S fuzzy system. *International Journal of Robust and Nonlinear Control* 30(4): 1622–1642.
- Cao K, Wang B, Ding H, Lv L, Tian J, Hu H, Gong F (2021) Achieving reliable and secure communications in wireless-powered NOMA systems. *IEEE Trans Veh Technol* 70(2):1978–1983
- Chen Y (2022) Research on collaborative innovation of key common technologies in new energy vehicle industry based on digital twin technology. *Energy Rep* 8:15399–15407
- Cheng B, Wang M, Zhao S, Zhai Z, Zhu D, Chen J (2017) Situation-aware dynamic service coordination in an IoT environment. *IEEE/ACM Trans Netw* 25(4):2082–2095
- Cheng L, Yin F, Theodoridis S, Chatzis S, Chang TH (2022) Rethinking Bayesian learning for data analysis: The art of prior and inference in sparsity-aware modeling. *IEEE Signal Process Mag* 39(6):18–52
- Hazrat B, Yin B, Kumar A, Ali M, Zhang J, Yao J (2023) Jerk-bounded trajectory planning for rotary flexible joint manipulator: an experimental approach. *Soft Comput* 27(7):4029–4039. <https://doi.org/10.1007/s00500-023-07923-5>
- Jiang S, Zhao C, Zhu Y, Wang C, Du Y, (2022) A practical and economical ultra-wideband base station placement approach for indoor autonomous driving systems. *J Adv Trans* 2022.
- Kumar A, Shaikh AM, Li Y et al (2021) Pruning filters with L1-norm and capped L1-norm for CNN compression. *Appl Intell* 51:1152–1160. <https://doi.org/10.1007/s10489-020-01894-y>
- Li Y, Qian J, Feng S, Chen Q, Zuo C (2022) Deep-learning-enabled dual-frequency composite fringe projection profilometry for single-shot absolute 3D shape measurement. *Opto-Electron Adv* 5(5):210021–210031
- Li L, Wang P, Zheng X, Xie Q, Tao X, Velásquez JD (2023) ual-interactive fusion for code-mixed deep representation learning in tag recommendation. *Information Fusion*, p 101862.
- Liu AA, Zhai Y, Xu N, Nie W, Li W, Zhang Y (2021) Region-aware image captioning via interaction learning. *IEEE Trans Circuits Syst Video Technol* 32(6):3685–3696
- Liu Y, Wang K, Liu L, Lan H, Lin L (2022a) Tcgl: temporal contrastive graph for self-supervised video representation learning. *IEEE Trans Image Process* 31:1978–1993
- Liu D, Cao Z, Jiang H, Zhou S, Xiao Z, Zeng F (2022b) Concurrent low-power listening: a new design paradigm for duty-cycling communication. *ACM Trans Sens Netw* 19(1):1–24
- Liu H, Xu Y, Chen F (2023a) Sketch2Photo: Synthesizing photo-realistic images from sketches via global contexts. *Eng Appl Artif Intell* 117:105608
- Liu X, Shi T, Zhou G, Liu M, Yin Z, Yin L, Zheng W (2023b) Emotion classification for short texts: an improved multi-label method. *Human Soc Sci Commun* 10(1):1–9
- Lu Z, Cheng R, Jin Y, Tan KC, Deb K (2023) Neural architecture search as multi-objective optimization benchmarks: Problem formulation and performance assessment. *IEEE transactions on evolutionary computation*.
- Luo R, Peng Z, Hu J (2023) On model identification based optimal control and its applications to multi-agent learning and control. *Mathematics* 11(4):906
- Lv Z, Chen D, Feng H, Wei W, Lv H (2022) Artificial intelligence in underwater digital twin's sensor networks. *ACM Trans Sens Netw (TOSN)* 18(3):1–27

- Mi C, Huang S, Zhang Y, Zhang Z, Postolache O (2022) Design and implementation of a 3-D measurement method for container handling target. *J Marine Sci Eng* 10(12):1961
- Muhammad IQ, Majid A, Shamrooz S (2023) Adaptive event-triggered robust H_∞ control for Takagi–Sugeno fuzzy networked Markov jump systems with time-varying delay. *Asian J Control* 25(1): 213–228
- Ni Q, Guo J, Wu W, Wang H (2022) Influence-based community partition with sandwich method for social networks. *IEEE Transactions on Computational Social Systems*, 1–12
- Shamrooz M, Li Q, Hou J (2021) Fault detection for asynchronous T-S fuzzy networked Markov jump systems with new event-triggered scheme. *IET Control Theory Appl* 15(11):1461–1473
- Shen Y, Ding N, Zheng HT, Li Y, Yang M (2020) Modeling relation paths for knowledge graph completion. *IEEE Trans Knowl Data Eng* 33(11):3607–3617
- Ullah R, Dai X, Sheng A (2020) Event-triggered scheme for fault detection and isolation of non-linear system with time-varying delay. *IET Control Theory Appl* 14(16):2429–2438
- Wang Y, Xu N, Liu AA, Li W, Zhang Y (2021) High-order interaction learning for image captioning. *IEEE Trans Circuits Syst Video Technol* 32(7):4417–4430
- Wang Q, Hu J, Wu Y, Zhao Y (2023a) Output synchronization of wide-area heterogeneous multi-agent systems over intermittent clustered networks. *Inf Sci* 619:263–275
- Wang Z, Zhao D, Guan Y (2023b) Flexible-constrained time-variant hybrid reliability-based design optimization. *Struct Multidiscip Optim* 66(4):89
- Wang L, Zhai Q, Yin B et al (2019) Second-order convolutional network for crowd counting. In: *Proc. SPIE* 11198, fourth international workshop on pattern recognition, 111980T; <https://doi.org/10.1117/12.2540362>.
- Xie X, Xie B, Cheng J, Chu Q, Dooling T (2021) A simple Monte Carlo method for estimating the chance of a cyclone impact. *Nat Hazards* 107:2573–2582
- Xie X, Tian Y, Wei G (2023a) Deduction of sudden rainstorm scenarios: integrating decision makers' emotions, dynamic Bayesian network and DS evidence theory. *Nat Hazards* 116(3):2935–2955
- Xie X, Huang L, Marson SM, Wei G (2023b) Emergency response process for a sudden rainstorm and flooding: scenario deduction and Bayesian network analysis using evidence theory and knowledge meta-theory. *Nat Hazards* 117(3):3307–3329
- Xu H, Sun Z, Cao Y et al (2023) A data-driven approach for intrusion and anomaly detection using automated machine learning for the Internet of Things. *Soft Comput.* <https://doi.org/10.1007/s00500-023-09037-4>
- Yang S, Li Q, Li W, Li X, Liu AA (2022) Dual-level representation enhancement on characteristic and context for image-text retrieval. *IEEE Trans Circuits Syst Video Technol* 32(11):8037–8050
- Yao W, Guo Y, Wu Y, Guo J (2017) Experimental validation of fuzzy PID control of flexible joint system in presence of uncertainties. In: *2017 36th Chinese Control Conference (CCC)* (pp 4192–4197). IEEE. <https://doi.org/10.23919/ChiCC.2017.8028015>.
- Yao Y, Shu F, Li Z, Cheng X, Wu L (2023) Secure transmission scheme based on joint radar and communication in mobile vehicular networks. *IEEE transactions on intelligent transportation systems*
- Yin B, Aslam MS et al (2023) A practical study of active disturbance rejection control for rotary flexible joint robot manipulator. *Soft Comput* 27:4987–5001. <https://doi.org/10.1007/s00500-023-08026-x>
- Yin B, Khan J, Wang L, Zhang J, Kumar A (2019) Real-time lane detection and tracking for advanced driver assistance systems. In: *2019 Chinese Control Conference (CCC)* (pp 6772–6777). IEEE. <https://doi.org/10.23919/ChiCC.2019.8866334>.
- Yuan H, Yang B (2022) System dynamics approach for evaluating the interconnection performance of cross-border transport infrastructure. *J Manag Eng* 38(3):04022008
- Zhang X, Wang Z, Lu Z (2022a) Multi-objective load dispatch for electric vehicles using modified gravitational search and Particle Swarm Optimization algorithm. *Appl Energy* 306:118018
- Zhang J, Tang Y, Wang H, Xu K (2022b) ASRO-DIO: Active subspace random optimization-based depth inertial odometry. *IEEE Trans Rob* 39(2):1496–1508
- Zhang J, Peng S, Gao Y, Zhang Z, Hong Q (2023) APMSA: adversarial perturbation against model stealing attacks. *IEEE Trans Inf Forensics Secur* 18:1667–1679
- Zhao K, Jia Z, Jia F, Shao H (2023) Multi-scale integrated deep self-attention network for predicting the remaining useful life of aero-engine. *Eng Appl Artif Intell* 120:105860
- Zhou X, Sun K, Wang J, Zhao J, Feng C, Yang Y, Zhou W (2023) Computer vision enabled building digital twin using building information model. *IEEE Trans Industr Inf* 19(3):2684–2692

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